Girls Build Bots — Cadette Digital Game Design

Badge-aligned workshop guide for robotics-team-led outreach

Level: Cadettes (Grades 6–8) · **Badge:** Coding for Good — Digital Game Design **Series:** Coding for Good (Coding Basics · Digital Game Design · App Development) **Format:** One 2-hour session · **Ratio:** 1 mentor per 5–6 Scouts **Who runs it:** A FIRST robotics team (FLL / FTC / FRC) leading the session. A physical robot is a nice bonus but **not required** — every activity runs on free browser tools.

> **Troop Leaders:** This curriculum is designed to *align with* the Cadette Coding for Good badge series. Official badge requirements are in your Girl Scout Volunteer Toolkit (VTK). Verify completed activities against the official steps before awarding badges. Badges are purchased through your council shop.

The through-line: coding *for good*

Cadettes can handle the designer's real responsibility: games shape how players feel. Mechanics create emotions — tension, mastery, delight — and good designers wield that deliberately and ethically (fun without manipulation).

Vocabulary: mechanic, dynamic, difficulty curve, game feel, balance, playtest protocol, iteration, scope.

Digital Game Design (2 hours)

Badge theme: Designing and prototyping a digital game with real mechanics, structured playtesting, and iteration.

Materials

- Laptops with **MakeCode Arcade** (arcade.makecode.com)
- Design doc template: Core mechanic / Goal / Fail state / Difficulty curve / Intended feeling
- Playtest protocol sheets (observation half + interview half)
- Projector; sticky notes

Activity 1 — Mechanics Make Feelings (20 min)

Analyze how designers engineer emotion: a countdown timer creates *tension*; a growing combo counter creates *mastery*; a near-miss creates *"one more try."* Pick a game the room knows and dissect: what's its **core mechanic**, and what feeling does it manufacture? Add the ethics beat Cadettes are ready for: the same tools can be used to respect players or to hook them unfairly — designing *for good* means fun that leaves players glad they played.

Activity 2 — Design Doc with a Difficulty Curve (20 min)

Each Scout (or pair) writes a one-page design doc: one core mechanic, a goal, a **fail state**, the intended feeling, and — new at this level — a sketched **difficulty curve** (how does minute three get harder than minute one? faster enemies? shorter timer? more obstacles?). Scope rule: one mechanic, executed well, beats four executed poorly.

Activity 3 — Build the Prototype (50 min)

Build in MakeCode Arcade to each doc: 1. Core loop: player, objective, fail state 2. Score + win/lose conditions 3. **The difficulty curve, implemented** — a variable that ramps speed, spawn rate, or time pressure as score climbs 4. Game feel pass: add sound effects and a screen shake or flash on key events — tiny touches, huge difference 5. Stretch: a title screen and a "play again" flow

Activity 4 — Structured Playtest + Iterate (20 min) + Wrap (10 min)

Run a real **playtest protocol**, not just a swap: the designer silently *observes* a tester play for two minutes (where do they struggle? smile? quit?), then asks three interview questions (What was the goal? What frustrated you? What would you change?). Notes go on the protocol sheet; designer implements **one change** from the data. Closing: each designer names the observation that surprised them most — that surprise is why studios playtest.

> **Robot moment (optional):** Share for 60 seconds how the team dissects each season's new competition game — mechanics, scoring, fail states — before designing a robot to win it. Same analysis, 120-pound game piece.

Completion Checklist — Digital Game Design

By the end of the session, to earn the badge, each Scout can: - [ ] Explain how a game mechanic creates a specific player feeling - [ ] Write a design doc with a core mechanic, fail state, and difficulty curve - [ ] Build a playable prototype implementing that difficulty curve - [ ] Run a structured playtest and iterate based on observed data

---

Mentor Guide (for the robotics team)

- **Treat Cadettes as junior designers.** Use the real vocabulary; they'll rise to it.
- **You don't need a robot for this.** The "robot moment" is a 60-second add-on.
- **Guard scope like a producer.** One mechanic, polished. Every cut feature is a win — say so.
- **Teach silent observation.** Designers ache to explain their game mid-test; coach them to watch instead. What testers *do* outranks what they *say*.
- **Praise the game-feel pass.** The sound-and-shake polish minute is where prototypes start feeling like real games — let them enjoy it.

📣 Share Your Impact — required for teams and troops using this curriculum

If you use Girls Build Bots materials to run a workshop, please complete all three:

1. **Credit us.** Tell your Scouts and audience that the curriculum comes from Girls Build Bots, and credit "Girls Build Bots" anywhere you share about the event.
2. **Tag us.** Post about your workshop and tag **@girlsbuildbots** on Instagram.
3. **Report your results.** Email **girlsbuildbots@gmail.com** with: the number of girls who participated, the number of badges earned, and a sentence or two of feedback on how it went.

This is how the program grows and how we measure our impact — your report becomes part of the Girls Build Bots story and helps the next troop and team.

Created for the "Girls Build Bots" Girl Scout Gold Award project. Original curriculum aligned to the Cadette Coding for Good badge series; official requirements © Girl Scouts of the USA, available to troop leaders in the Volunteer Toolkit.